

ANNEX A

NATIONAL ENSO EARLY WARNING AND CLIMATE INTELLIGENCE SYSTEM (NEWCIS)

CONCEPT NOTE

1. OVERVIEW

The **National ENSO Early Warning and Climate Intelligence System (NEWCIS)** is proposed as Papua New Guinea's integrated national platform for monitoring, analysing, visualising, and reporting ENSO-related risks and impacts.

The system will support whole-of-government decision-making through the integration of climate science, risk intelligence, geospatial analytics, and sectoral preparedness information into a single National Climate Intelligence Operating Picture.

2. PURPOSE

NEWCIS will:

- Provide real-time monitoring of ENSO indicators.
- Strengthen national early warning capability.
- Support climate-security risk assessments.
- Improve preparedness and response coordination.
- Enable evidence-based decision-making.
- Support national resilience and risk governance.

3. STRATEGIC OBJECTIVES

- a. Integrate national and international climate data sources.
- b. Establish a centralized climate intelligence capability.
- c. Improve monitoring of provincial vulnerabilities.
- d. Support Government preparedness planning.
- e. Strengthen climate-security governance under the National Security Policy 2024–2029.

4. KEY CAPABILITIES

- Real-time ENSO monitoring.
- National and provincial risk dashboards.
- GIS-based vulnerability mapping.

- Early warning alert system.
- Strategic intelligence reporting.
- Scenario modelling and forecasting.
- Critical infrastructure monitoring.

5. DATA SOURCES

National Sources

- PNG National Weather Service
- National Disaster Centre
- National Security Agency
- Treasury
- Finance
- National Planning
- Health
- Agriculture
- Water PNG
- CEPA

International Sources

- NOAA
- NASA
- WMO
- Australian Bureau of Meteorology
- Japan Meteorological Agency

6. PLATFORM

Primary Analytics Platform: Microsoft Power BI

Spatial Intelligence Platform: ArcGIS Enterprise

Outputs:

- Weekly ENSO Situation Reports
- Monthly Intelligence Assessments
- Executive Dashboards
- NSAC Briefings
- NEC Briefings

7. IMPLEMENTATION APPROACH

Phase 1 (2026) Establish ENSO Intelligence Cell and Dashboard.

Phase 2 (2027) Develop National Climate Intelligence Portal.

Phase 3 (2028–2030) Institutionalize National Climate Intelligence Centre.

8. EXPECTED OUTCOMES

- Improved national preparedness.
- Enhanced climate intelligence capability.
- Faster policy response.
- Better protection of vulnerable communities.
- Stronger whole-of-government coordination.